

Lecture 5: Costs of Production



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Lecture overview

- The Meaning of Costs
- Production in the Short run
- Costs in the Short run
- Production in the Long run
- Costs in the Long run

Opportunity Cost

- The cost of any activity measured in terms of the sacrifice made, specifically, the next best alternative forgone.
- Economists focus on what you give up, not just what you spend.
 - A factory can produce either 10 small cars or 6 large cars using the same inputs.
 - The Opportunity Cost: Producing 1 small car costs 0.6 of a large car.

Measuring a Firm's Costs

To determine true economic cost, factors of production are split into two categories:

1. Explicit Costs (Factors NOT owned by the firm)

- Direct monetary payments for resources.
 - Example: Paying £100 for electricity.
 - That £100 is no longer available to be spent on other business needs.

2. Implicit Costs (Factors ALREADY owned by the firm)

- Costs that do not involve a direct cash outlay but represent lost potential income.
 - Example: Owned Buildings: The opportunity cost is the rental income lost by not leasing the space to another firm. Labor of Owner: The salary the owner could have earned working elsewhere.

The Irrelevant Costs

- **Sunk Costs (Historic Cost):** What was originally paid for a machine is irrelevant; that money is gone and cannot be recovered by choosing not to use the machine.
- **Replacement Cost:** This only matters when deciding to buy a new machine, not when calculating the cost of using the current one.

Q Assume that a firm already owns a machine with a total life of 10 years. The cost of using the machine for one year to produce good X is:

- A. a tenth of what the firm paid for the machine in the first place.
- B. a tenth of what it would cost to replace the machine.
- C. the value of one year's output of X.
- D. the scrap value of the machine at the end of its life.
- E.

 the maximum the machine could have earned for the firm in some alternative use during the year in question.

- The economic cost is the **opportunity cost** of using the machine for good X instead of something else.
- Not a tenth of what the firm originally paid, that's historic cost, and it's sunk.
- Not a tenth of replacement cost, that matters only if buying a new machine.
- Not the value of output, that's revenue, not cost.
- Not just scrap value, scrap value may matter at end of life, but it's not the annual opportunity cost of use.

Production in the Short run

Production in the Short run

The cost of any output level is determined by:

- The quantity of inputs used.
- The price per unit of those inputs.

Fixed vs. Variable Factors:

- Fixed Factors: Inputs that cannot be increased within a specific timeframe (e.g., buildings, heavy machinery).
- Variable Factors: Inputs that can be changed quickly to alter output (e.g., raw materials, fuel, electricity, and labor/overtime).

Production in the Short run

The Time Horizons:

- Short Run: A period where at least one factor of production is fixed. Output can only grow by adding more variable factors.
- Long Run: A period long enough that all inputs can be varied.

The short run isn't a set number of days; it varies by industry (e.g., 1 year for a farm vs. 3 years for a shipping company).

Reminder: factors of production

- Land and raw material
- Labour
- Capital
- Entrepreneurship

The Short-Run Production Function

- The Production Function: A mathematical or graphical relationship showing how much output (TPP) is produced from specific inputs.

$$TPP = f(L_n, L_b)$$

Where L_n = Fixed Land and L_b = Variable Labor.

- The Law of Diminishing Marginal Returns: As more units of a variable factor (labor) are added to a fixed factor (land), the extra output gained from each new worker will eventually begin to decline.
 - Imagine a small office; adding a second person helps, but adding 50 people makes it impossible for anyone to work efficiently.

Short-Run Production

To analyze efficiency, economists look at two additional measures of the production function:

- Average Physical Product (APP): The output produced per unit of the variable factor (e.g., wheat per worker).

$$APP = \frac{TPP}{Q_v}$$

- Example: If 4 workers produce 36 tones, $APP = 36 / 4 = 9$ tones per worker.

Short-Run Production

- Marginal Physical Product (MPP): The extra output generated by employing one additional unit of the variable factor.

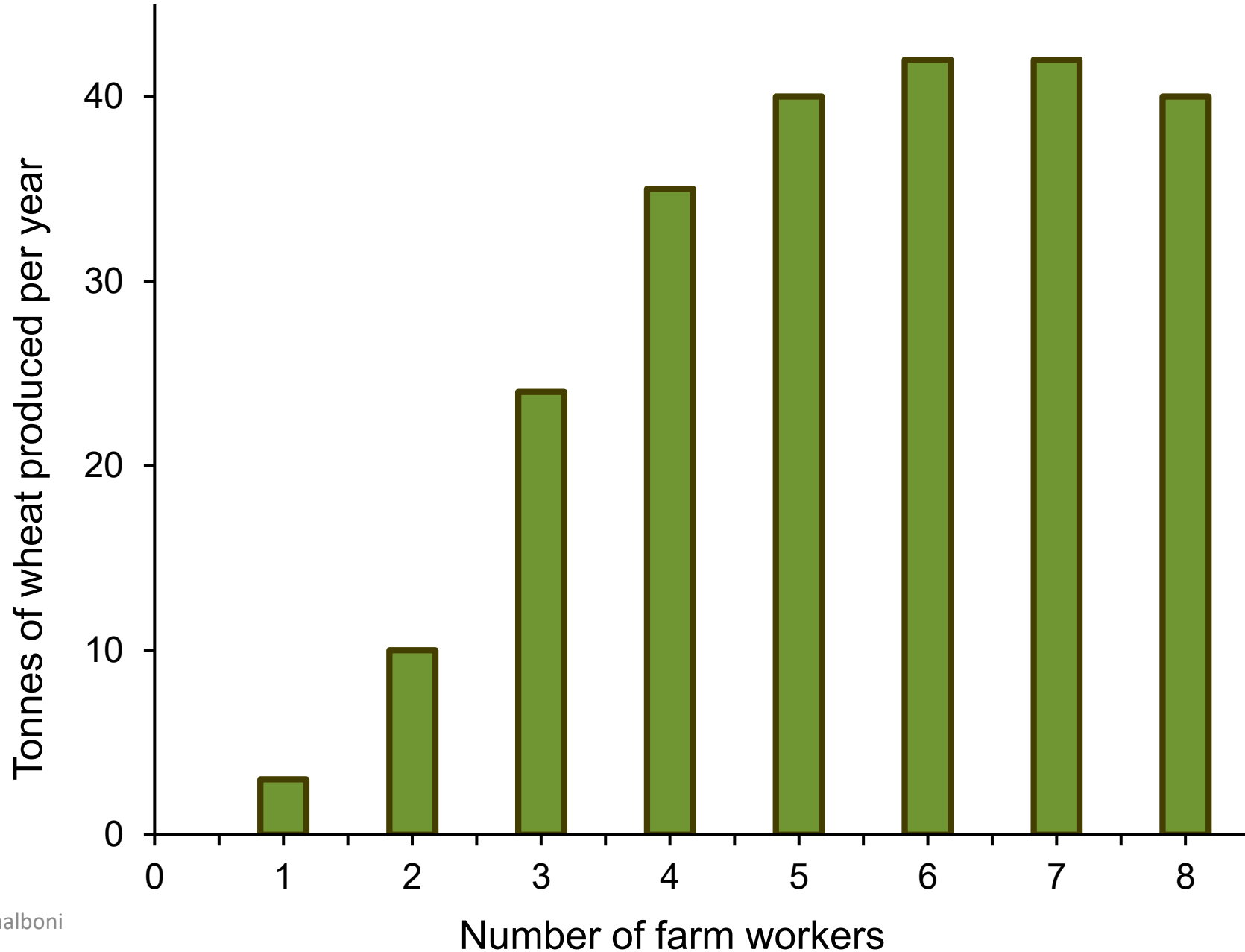
$$MPP = \frac{\Delta TPP}{\Delta Q_v}$$

- Example: If adding a 4th worker increases total output from 24 to 36 tones, the MPP of that worker is 12 tones.

Wheat production per year from a particular farm

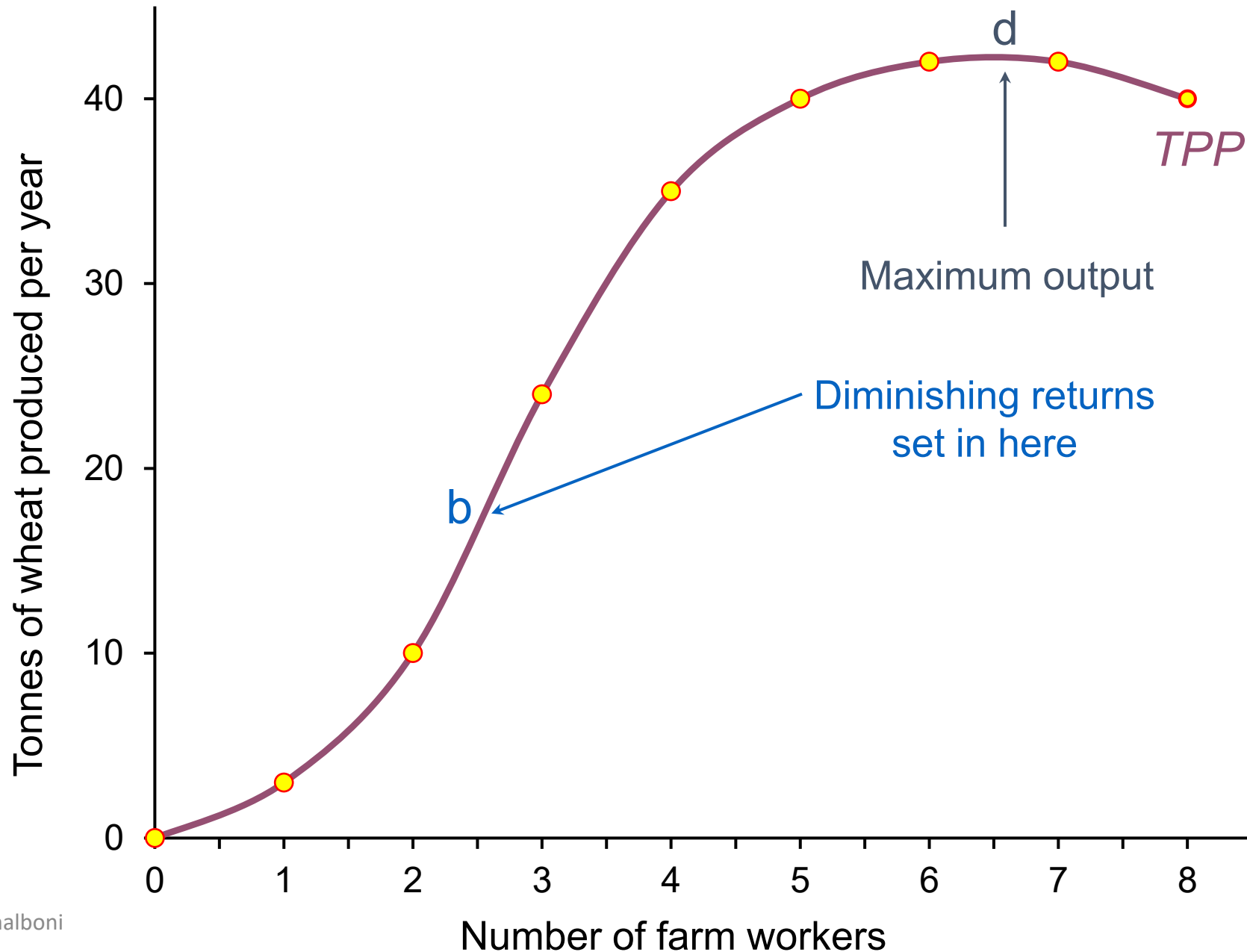
<u>Number of workers</u>	<u>TPP</u>	<u>APP</u>	<u>MPP</u>
0	0	0	
			3
1	3	3	
			7
2	10	5	
			14
3	24	8	
			12
4	36	9	
			4
5	40	8	
			2
6	42	7	
			0
7	42	6	
			-2
8	40	5	

Wheat production per year from a particular farm



<u>Number of workers</u>	<u>TPP</u>
0	0
1	3
2	10
3	24
4	36
5	40
6	42
7	42
8	40

Wheat production per year from a particular farm



Number of workers	<i>TPP</i>
0	0
1	3
2	10
3	24
4	36
5	40
6	42
7	42
8	40

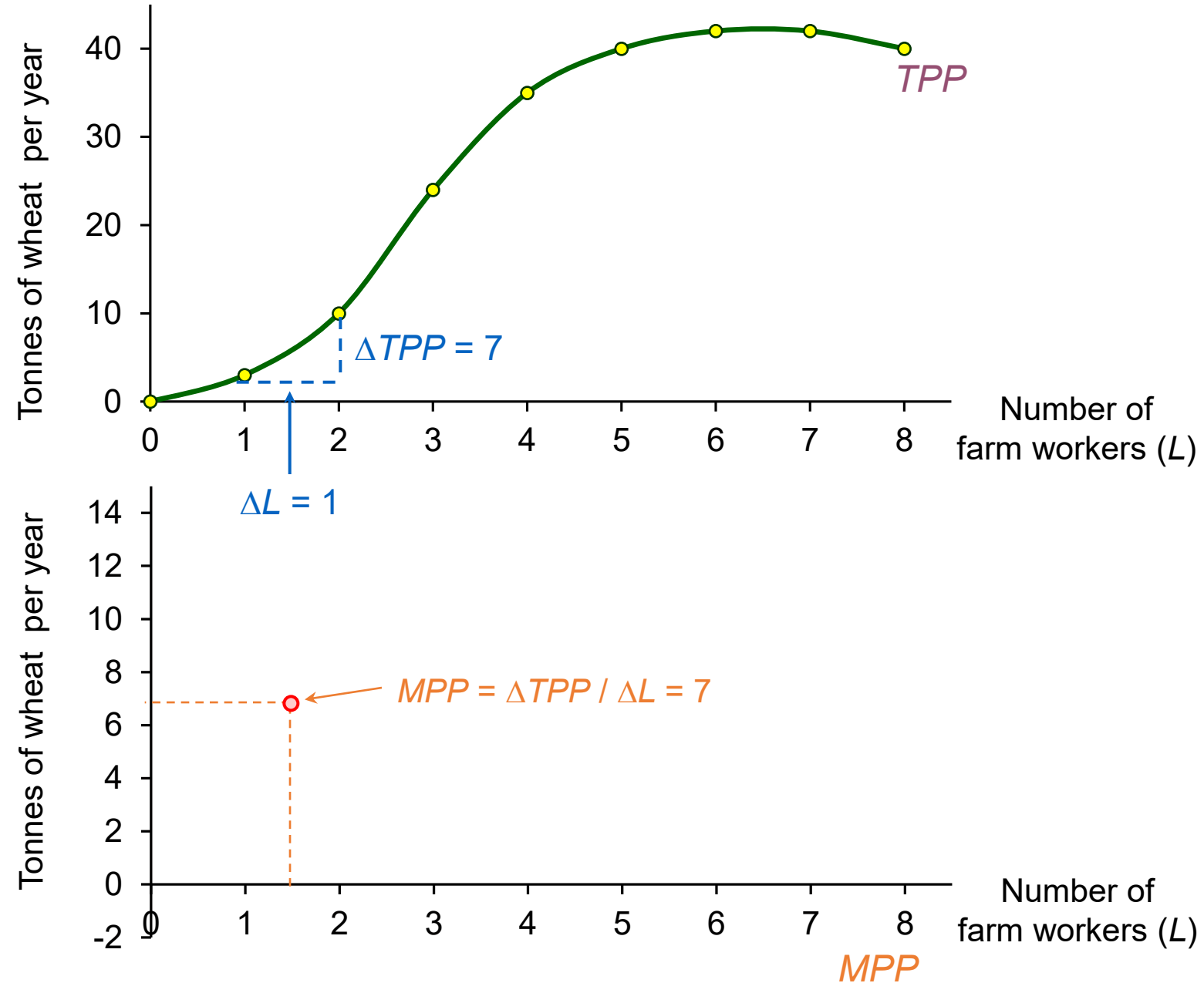
Total Physical Product (TPP) Curve

- Phase 1: Increasing Returns (Points a to b)
 - Initially, output rises rapidly.
 - Reason: Specialization and teamwork. A few workers are more efficient together than one worker trying to do everything alone.
- Phase 2: Diminishing Returns (After Point b)
 - The curve starts to flatten; output is still growing, but at a slower rate.
 - Reason: The fixed factor (land/machinery) is becoming crowded.
- Phase 3: Maximum Yield (Point d)
 - Output reaches its absolute peak. The fixed factor is being used to its full capacity.

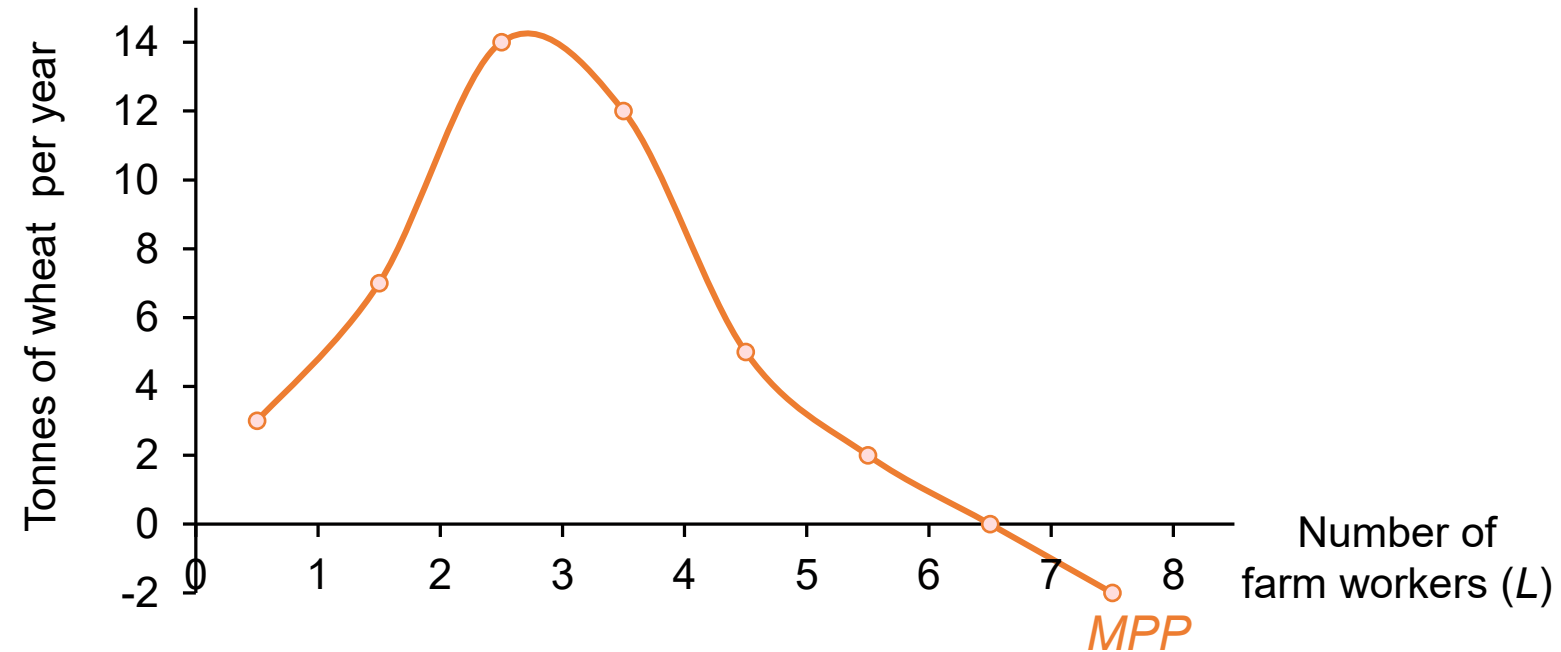
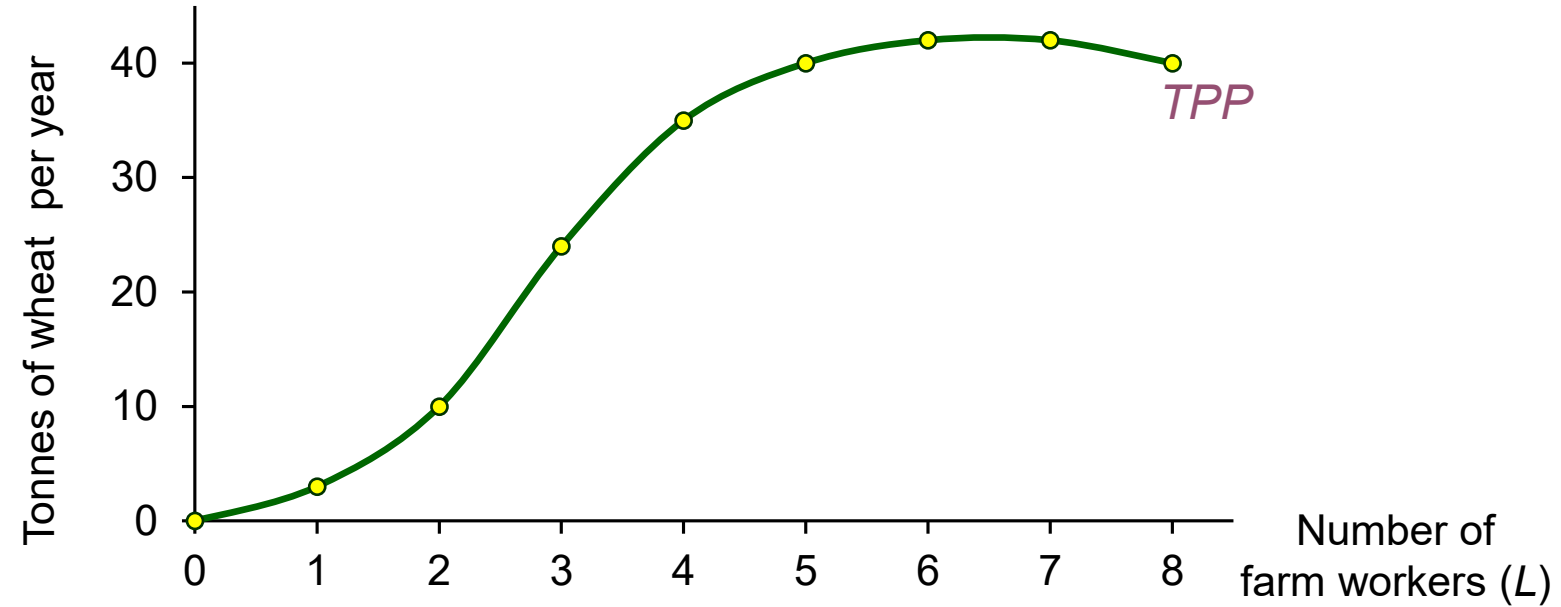
Negative Returns and Overcrowding

- Beyond Point d: Total output actually begins to fall.
- The "Too Many Cooks" Effect: Adding more workers (e.g., an 8th worker) becomes counterproductive.
 - Workers may physically get in each other's way or disrupt the workflow.
- Using more labor does not always equal more profit or more product; efficiency depends on the balance between fixed and variable inputs.

Wheat production per year from a particular farm



Wheat production per year from a particular farm



When Does Production Stop Growing?

The behavior of MPP dictates the direction of Total Product:

- Growth Phase: As long as MPP is positive (> 0), TPP will continue to rise. Even if the extra output is small, it is still an addition.
- The Maximum (Point d): When TPP is at its absolute maximum, the slope is zero. Therefore, $MPP = 0$. The next worker adds nothing.
- Negative Returns: Beyond Point d, TPP begins to fall. At this stage, MPP becomes negative.
 - Adding more workers at this stage actually reduces total wheat harvested (likely due to overcrowding or interference).

Costs in the Short run

Costs in the Short run

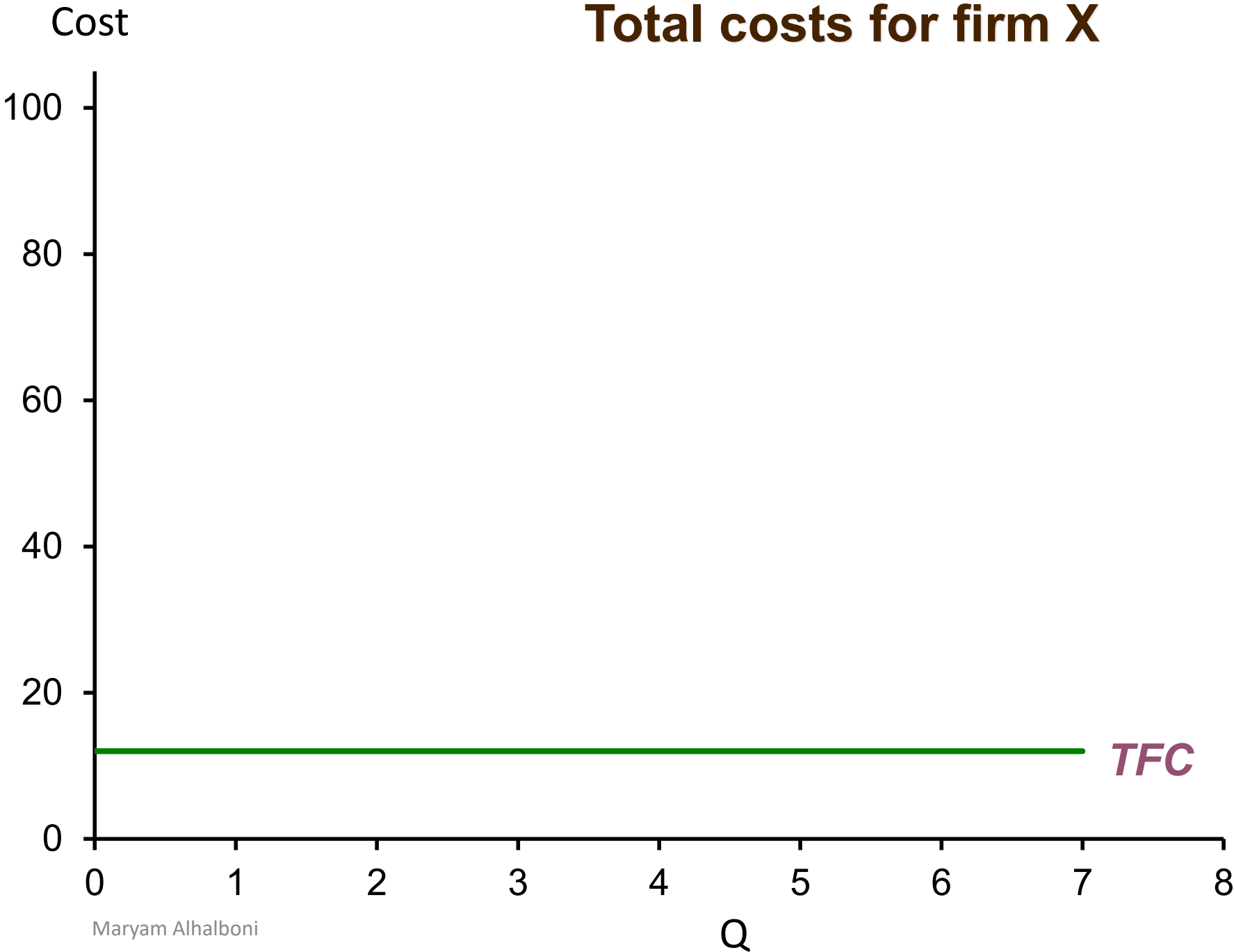
Two Primary Determinants:

- Factor Productivity: Higher productivity (TPP, APP, MPP) means fewer inputs are needed, leading to lower costs.
- Factor Price: The higher the price of inputs (wages, rent, etc.), the higher the production costs.

Classifying Short-Run Costs:

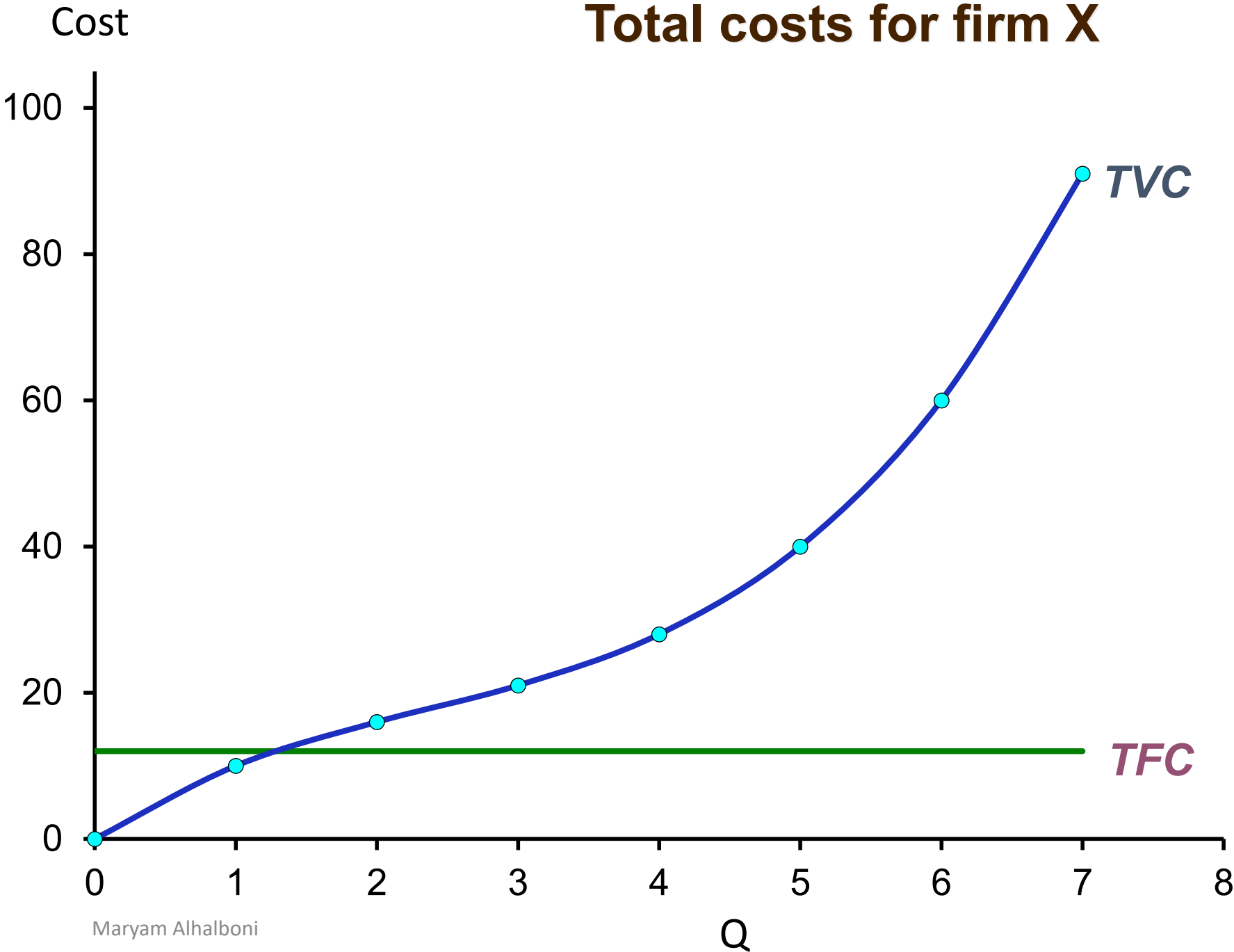
- Fixed Costs (TFC): Do not vary with output (e.g., rent on land). These must be paid even if production is zero.
- Variable Costs (TVC): Change directly with output (e.g., raw materials, fuel). More production = higher TVC.
- Total Cost (TC): $TC = TFC + TVC$

Total costs for firm X



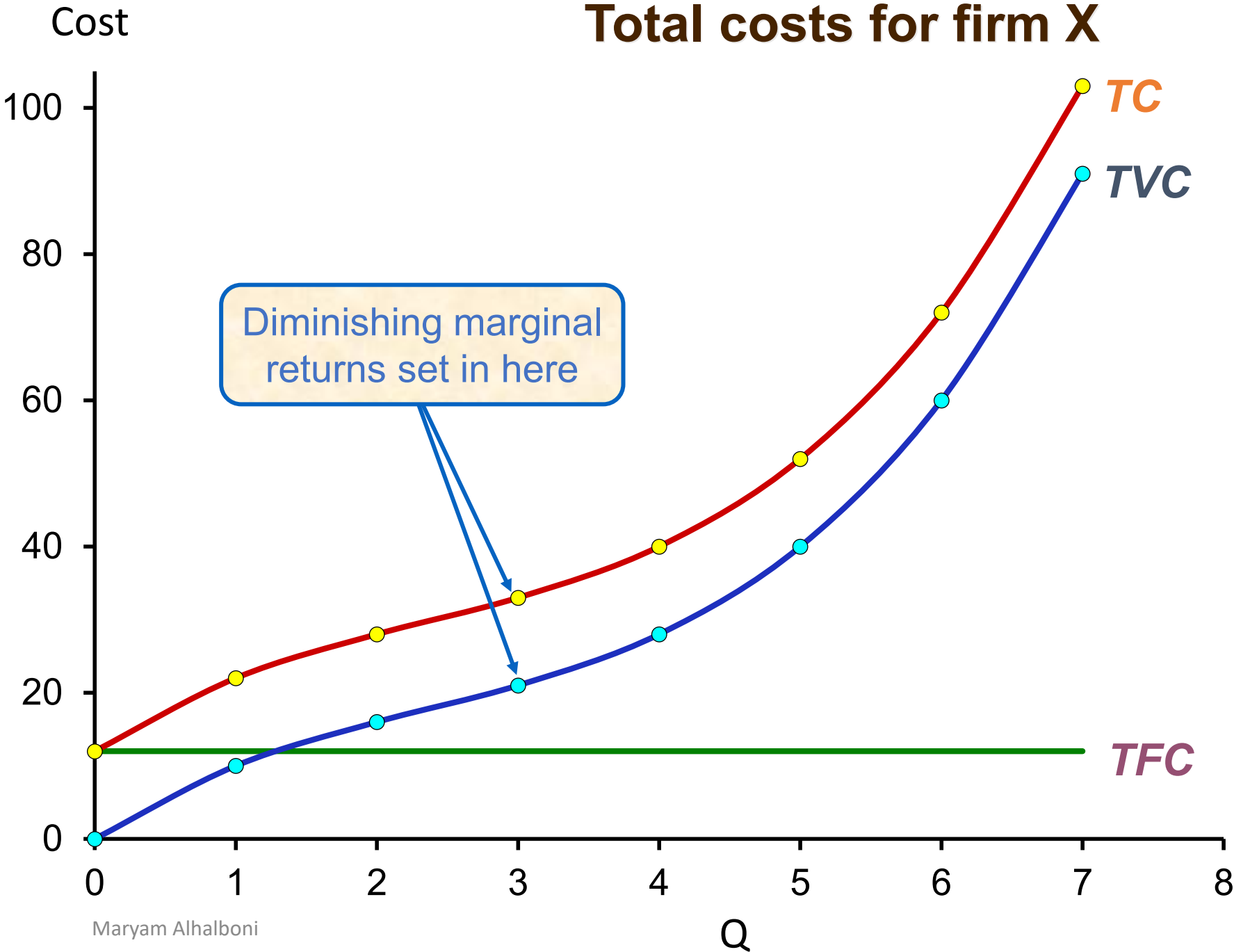
Output (Q)	TFC (£)
0	12
1	12
2	12
3	12
4	12
5	12
6	12
7	12

Total costs for firm X



Output (Q)	TFC (£)	TVC (£)
0	12	0
1	12	10
2	12	16
3	12	21
4	12	28
5	12	40
6	12	60
7	12	91

Total costs for firm X



Output (Q)	TFC (£)	TVC (£)	TC (£)
0	12	0	12
1	12	10	22
2	12	16	28
3	12	21	33
4	12	28	40
5	12	40	52
6	12	60	72
7	12	91	103

Total Cost Curves

- Total Fixed Cost (TFC): Represented as a horizontal line (e.g., at £12) because it remains constant regardless of output.
- Total Variable Cost (TVC): Starts at zero (no production = no variable cost).
 - Initial Phase: Rises slowly due to increasing returns and specialization (efficiencies).
 - Later Phase: Rises rapidly as Diminishing Returns set in. Extra workers produce less additional output, making each new unit more expensive to create.
- The TC curve is identical in shape to the TVC curve but is shifted upward by the amount of the Fixed Costs (£12).
- The vertical distance between TC and TVC is always constant, representing the Fixed Cost.

Q Which of the following is a fixed cost for a handmade furniture manufacturer?

- A. The cost of wood
- B. Labour costs
- ☒ C. The rent on the workshop
- D. The cost of glue and screws
- E. The cost of electricity for running the machines

Fixed Cost

- Because the firm pays rent whether it produces 0 tables or 50 tables.
- The others are generally variable:
 - wood
 - glue and screws
 - electricity for running machines (in most cases, this rises with usage)
 - labour costs (if paid hourly or per output)

Average Costs

- Average Cost (AC): The cost per unit of production.

$$AC = \frac{TC}{Q}$$

$$\text{Or } AC = AFC + AVC$$

- Average Fixed Cost $AFC = \frac{TEC}{Q}$ (This always declines as output increases).
- Average Variable Cost (AVC): $\frac{TVC}{Q}$

Marginal Costs

- Marginal Cost (MC): The extra cost of producing one more unit.

$$MC = \frac{\Delta TC}{\Delta Q}$$

- All marginal costs are variable costs; by definition, fixed costs do not change when one more unit is made.

Total, average and marginal cost for firm X

Output (Q) (units)	TFC (£)	AFC (TFC/Q) (£)	TVC (£)	AVC (TVC/Q) (£)	TC (TFC+TVC) (£)	AC (TC/Q) (£)	MC ($\Delta TC/\Delta Q$) (£)
0	12	—	0	—	12	—	
							10
1	12	12	10	10	22	22	
							6
2	12	6	16	8	28	14	
							5
3	12	4	21	7	33	11	
							7
4	12	3	28	7	40	10	
							12
5	12	2.4	40	8	52	10.4	
							20
6	12	2	60	10	72	12	
							31
7	12	1.7	91	13	103	14.7	

Production in the Long run

Production in the Long run

- In the long run, all factors of production are variable. Firms can build new factories, change technology, and alter the scale of operations.
- The Scale of Production: Unlike the short run (where we add labor to fixed land), the long run looks at what happens when all inputs increase proportionally.

Production in the Long run

Three Returns to Scale Scenarios:

- Constant Returns to Scale: Inputs and output grow by the same %.
 - Example: 4% increase in costs leads to exactly a 4% increase in output. Unit cost remains £0.50.
- Increasing Returns to Scale: Output grows by a larger % than inputs.
 - Example: 4% increase in costs leads to a 10% increase in output. Unit cost falls from £0.50 to £0.47.
- Decreasing Returns to Scale: Output grows by a smaller % than inputs.
 - Example: 4% increase in costs leads to only a 2% increase in output. Unit cost rises from £0.50 to £0.51.

Economies of Scale (Why Bigger can be Better)

Falling average costs (LRAC) as the scale of production increases.

- Plant Economies (Specific to one factory):
 - Specialization & Division of Labor: Workers focus on simple, repetitive tasks; this reduces training time and eliminates switching time between jobs.
 - Indivisibilities: Certain inputs (like a combine harvester) have a minimum size. High output is required to make these expensive machines economical.
 - The Container Principle: Doubling the surface area of a container (cost) more than doubles its volume (capacity/output).

Economies of Scale (Why Bigger can be Better)

- Plant Economies (Specific to one factory):
 - Efficiency of Large Machines: One operator can often run a large machine just as easily as a small one, but with much higher output.
 - By-products: Waste materials (like sawdust) only become plentiful enough to turn into new products (like charcoal) at a large scale.
 - Multistage Production: Integrating several production steps in one factory (e.g., turning trees into boxes) saves transportation and handling costs.

Economies of Scale (Why Bigger can be Better)

- Firm-Level Economies (The whole organization):
 - Organisational & Managerial: Centralized administration and specialized functions (HR, Accounting) reduce redundancy.
 - Spreading Overheads: Massive fixed costs, such as Research & Development (R&D), are spread over a larger number of units.
 - Financial Economies: Large firms access cheaper credit/loans due to lower perceived risk.
 - Bulk Buying: Volume discounts are easier to negotiate because the supplier's opportunity cost of losing the account is higher.
 - Economies of Scope: Costs are shared across a range of different products (e.g., Sony making both TVs and consoles).

Q Economies of scale can be defined as large-scale production leading to:

- A. bigger profits.
- ☒ B. lower costs per unit of production.
- C. greater marginal productivity of factors.
- D. greater output per unit of input.
- E. a better organisation of the factors of production.

Diseconomies and External Factors

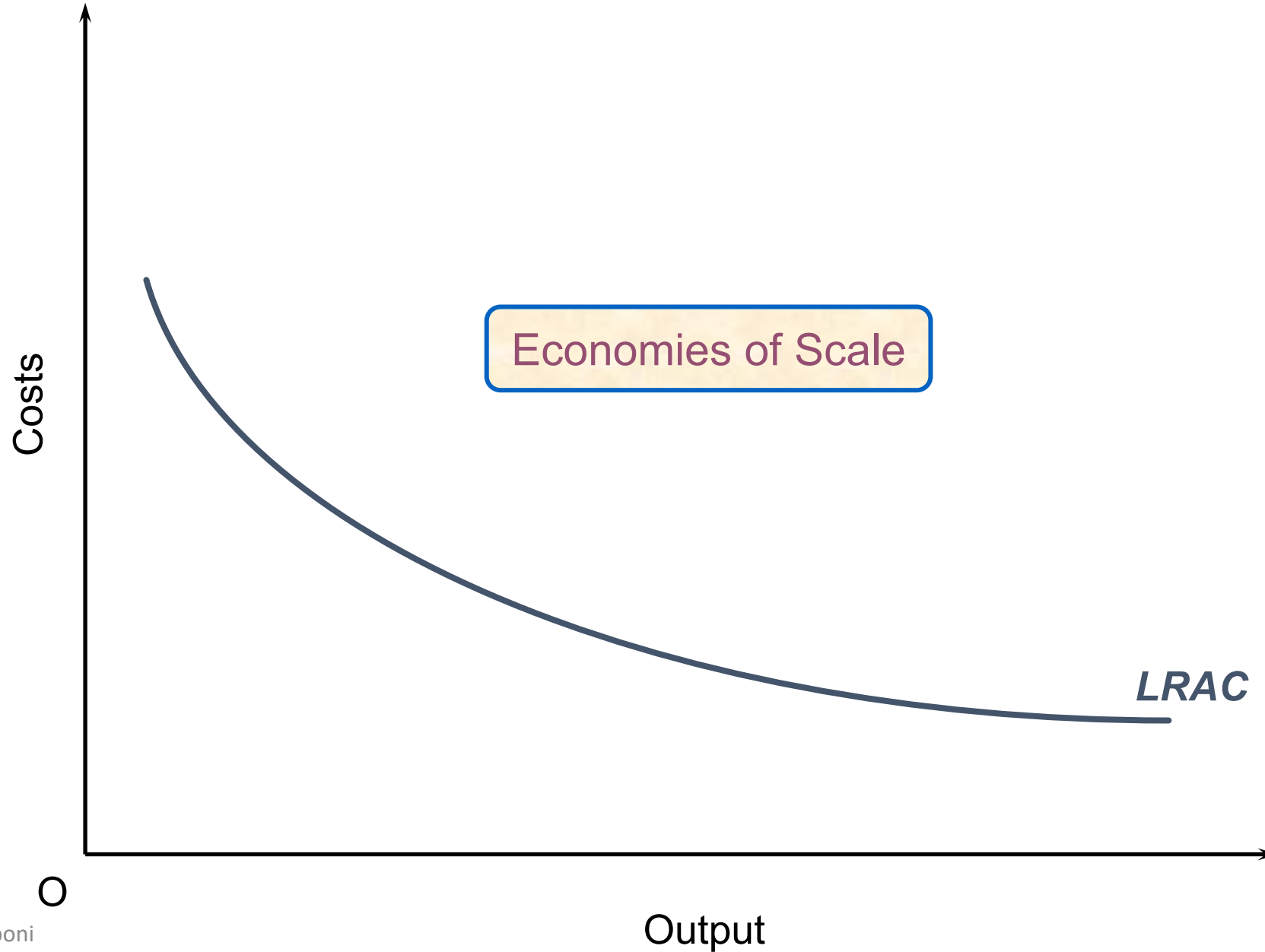
- Diseconomies of Scale (Internal): When a firm becomes too large, unit costs rise due to:
 - Management Issues: Poor communication, "red tape," and lack of personal oversight.
 - Worker Alienation: Staff feel like "cogs in a machine," leading to low motivation and poor industrial relations.
- External Factors (The whole industry):
 - External Economies: A firm benefits because the industry is large (e.g., a pool of local skilled labor or specialist suppliers).
 - External Diseconomies: Industry growth drives up prices for everyone (e.g., a shortage of raw materials or rising local land prices).

Costs in the Long run

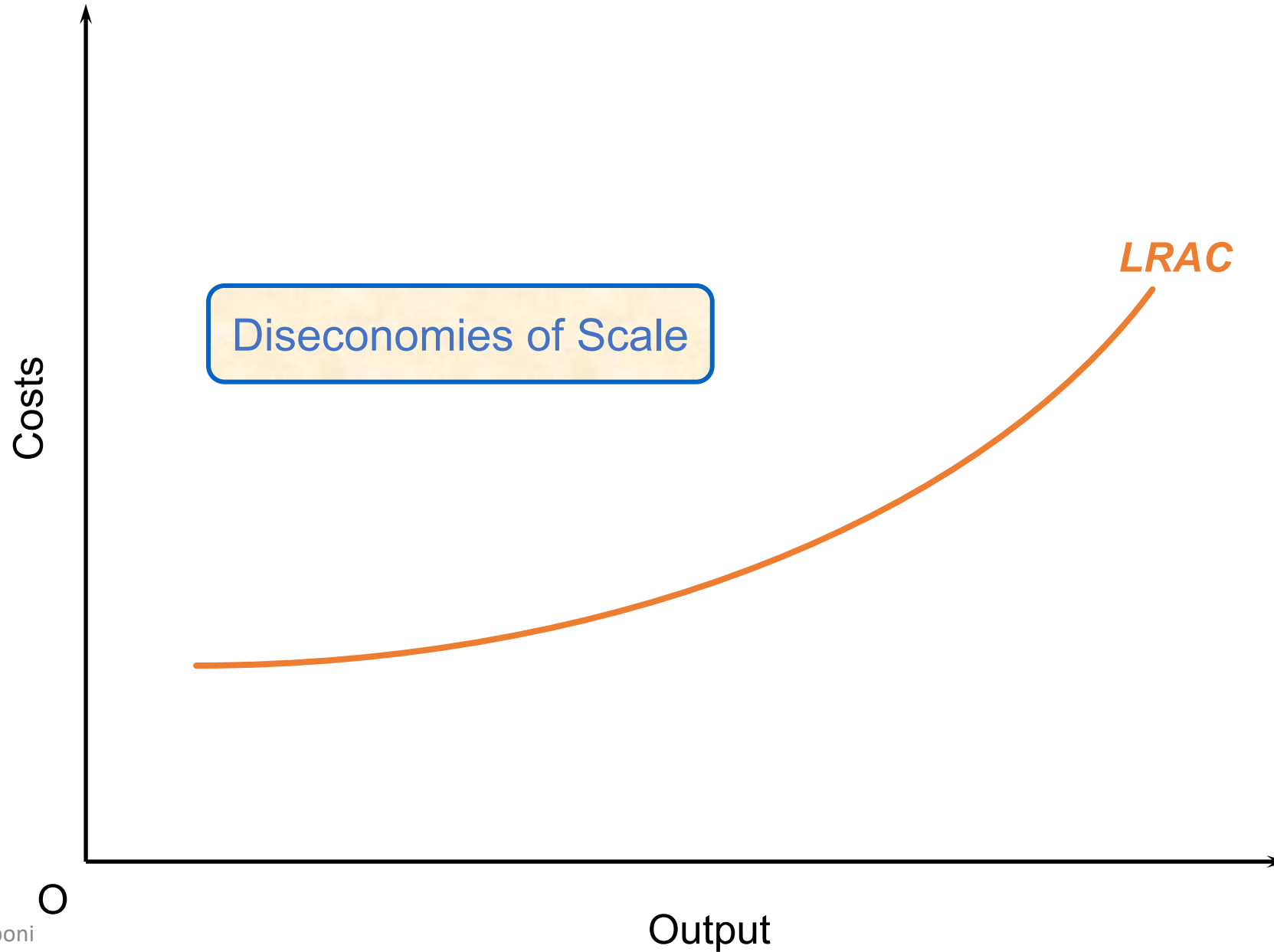
Long-Run Average Cost (LRAC)

- No Fixed Costs: Since everything can be changed, all long-run costs are variable.
- The Shape of the LRAC Curve: Often described as "U-shaped" or "L-shaped."
 - Downward Slope: Driven by economies of scale.
 - Flat Bottom (Q1 to Q2): A period of Constant Returns to Scale where the firm has reached the "Minimum Efficient Scale."
 - Upward Slope: Eventually, managerial diseconomies kick in, and unit costs begin to rise.
- Real-World Evidence: Most firms find it easy to achieve technical economies, but the "giant" firms often struggle with the human/managerial side of scale.

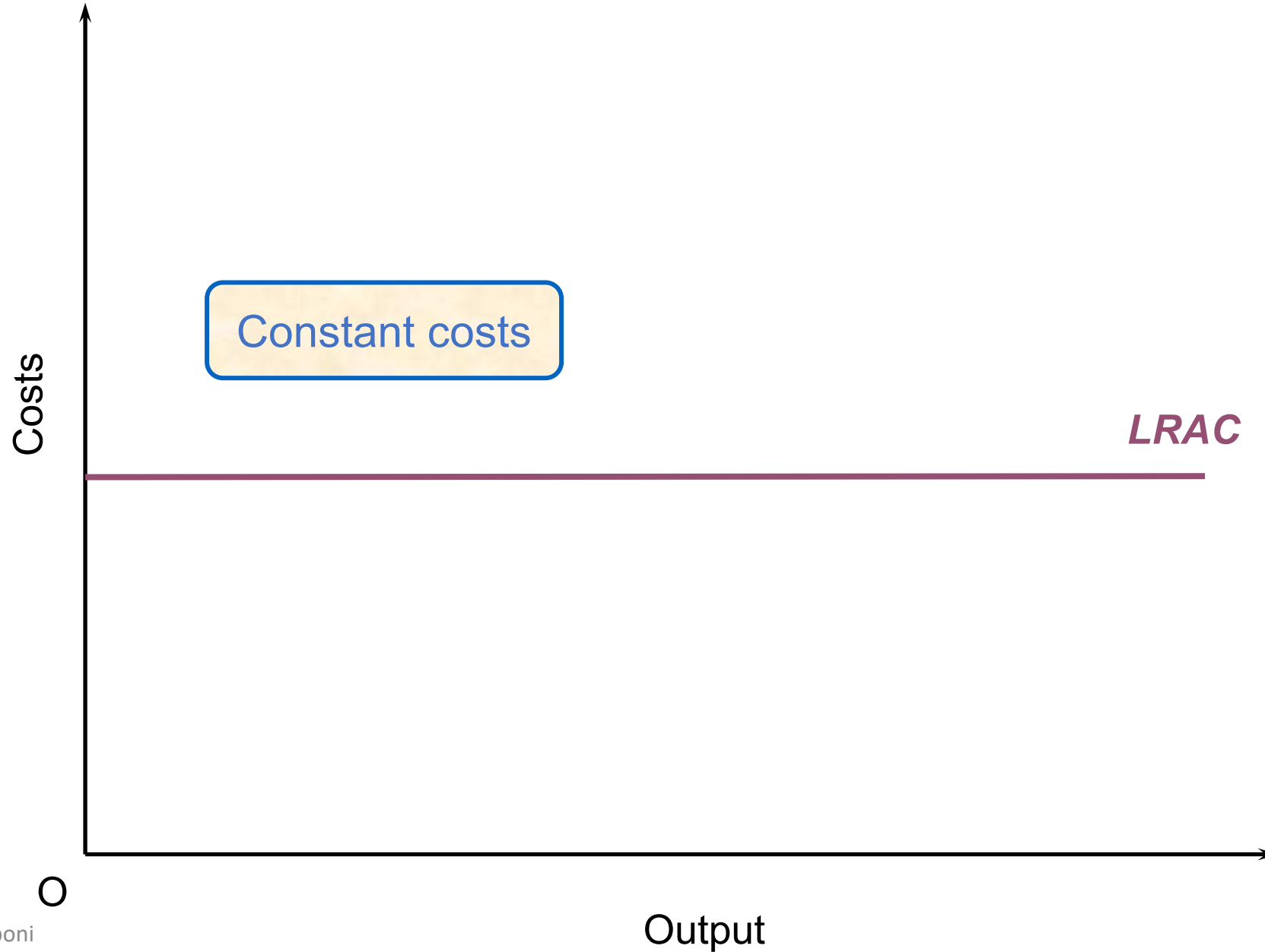
Alternative long-run average cost curves



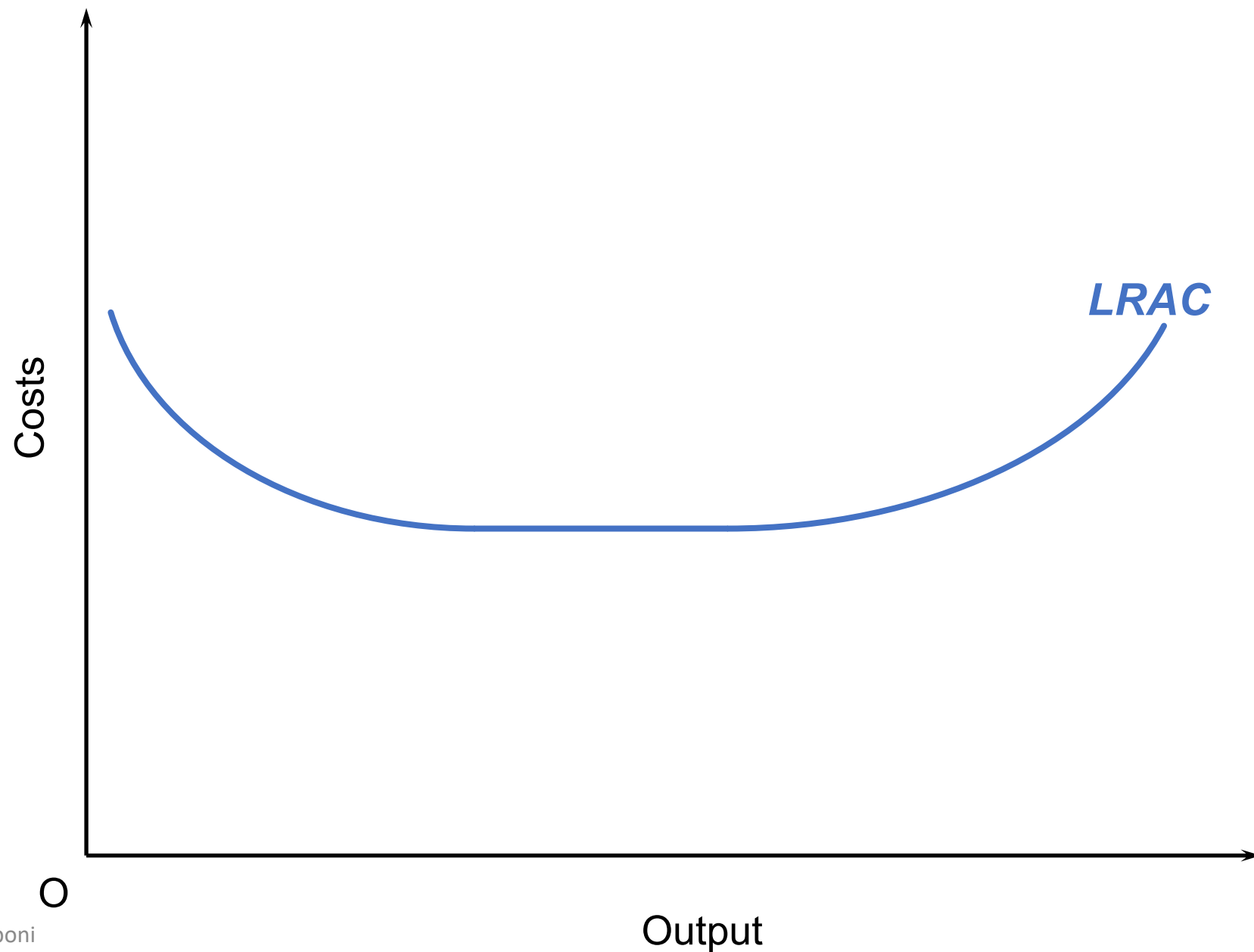
Alternative long-run average cost curves



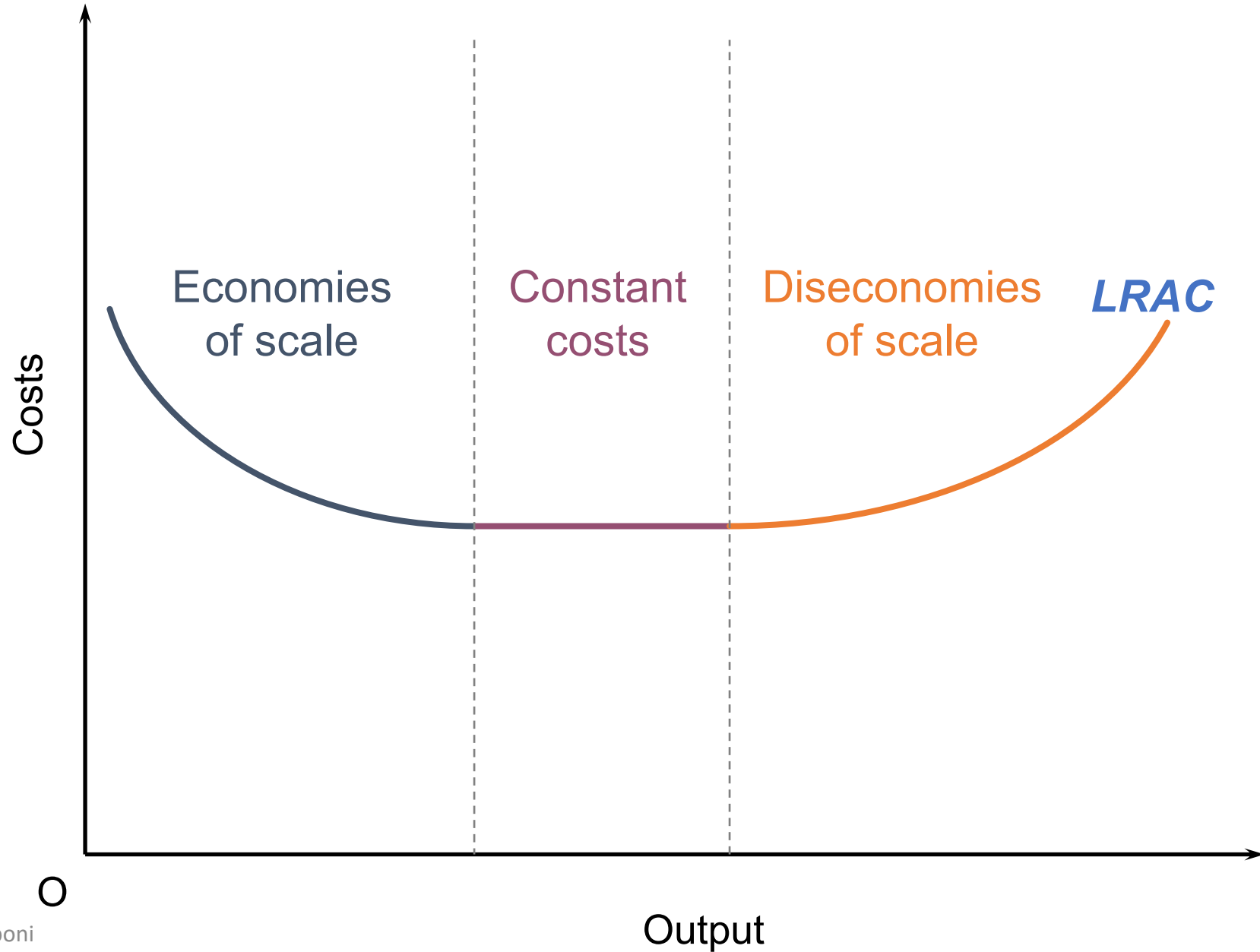
Alternative long-run average cost curves



A typical long-run average cost curve



A typical long-run average cost curve



Questions